General Linear Least Squares

input : data $(\vec{x}_i, y_i)$ for $i = 1, \ldots, n$; basis functions $\phi_1, \ldots, \phi_m$
output: $\theta^* = \vec{a} = (a_1, \ldots, a_m)$, coefficients of $\hat{y} = a_1\phi_1(\vec{x}) + \cdots + a_m\phi_m(\vec{x})$

for $i \leftarrow 1$ **to** n **do** `// build the` $n \times m$ `matrix` Z
  for $j \leftarrow 1$ **to** m **do**
    $Z_{ij} \leftarrow \phi_j(\vec{x}_i)$
  end
end

solve $\left(Z^T Z\right) \vec{a} = Z^T \vec{y}$ for $\vec{a}$ `// normal equations,` $m \times m$

return $\theta^* = \vec{a}$